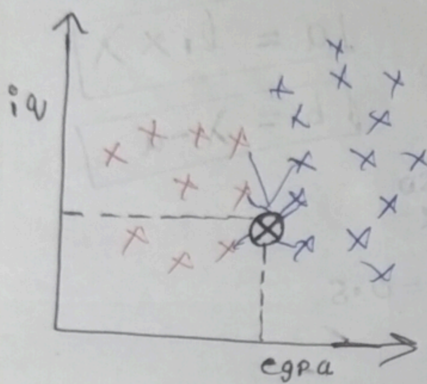
K-Nearest Neighbors (KNN) Classifier

KNN (K- Nearest Neighbors) is a Supervise Machine Learning algorithm uses for classification & Regression.
KNN predict the class label

→ In classification, KNN predict the class label of a new data point based on the most common class among its nearest neighbors.



→ class-1
→ class-0

- K → number of nearest data points considered
- N → based on nearest distance
- Neighbors → nearby data points in the feature space.

→ Non-Parametric :-

- KNN does not learn parameters like weights or coefficients. It stores the entire training dataset.

→ Lazy Learner :-

- No training happens during fit( )
- All computation happens during prediction

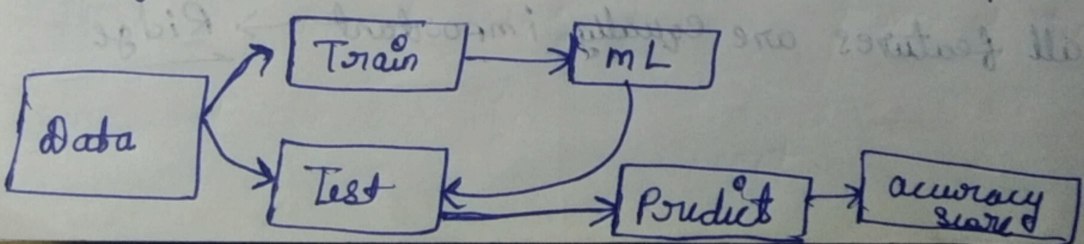
How KNN Works :-

- Choose a value of k (number of Neighbors)
- For a new data point
  → Calculate the distance b/w this point and all training points

→ select the k-Nearest points

→ performing majority voting among these k-points

→ Assign the class with the highest votes.



Importance Of feature Scaling :-

KNN is distance- based, so scale matters.

Example :- feature 1 : Age (0 – 100)
feature 2 : Salary (0 – 100000)

→ without scaling : Salary dominates distance
→ with scaling : All features contribute Equally

Choosing Value of K :-

i) Small (K=1)
- over sensitive to noise
- overfitting
- High Variance

Large K
ii) Smooth decision boundary
- underfitting
- High bias

data
→ heustric
→ Experimentation

heustric

$\sqrt{n} \Rightarrow$ n - no. of observation

Ex:- $\boxed{400} \Rightarrow \sqrt{400} = 20$

take odd value for K
$\boxed{19}$ $\boxed{21}$

if k = 4.
→ class1 = 2
→ class2 = 2

Experimentation

Cross validatation

n = 1000
→ 800 (Training)
→ 200 (Testing)

diff KNN → KNN-1 83
KNN-2 81
KNN-25 89 ✓

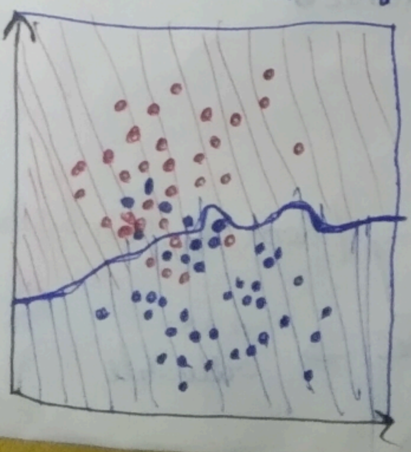
Trained again on Training data

Decision Surface [Decision Boundary] :-

A decision surface is the region in the feature space where a machine learning model decides which class a data point belongs to.

→ It separates different classes.

- for 2 features → It becomes decision boundary (line / curve)
- for more than 2 features → it becomes a surface or hyperplane.



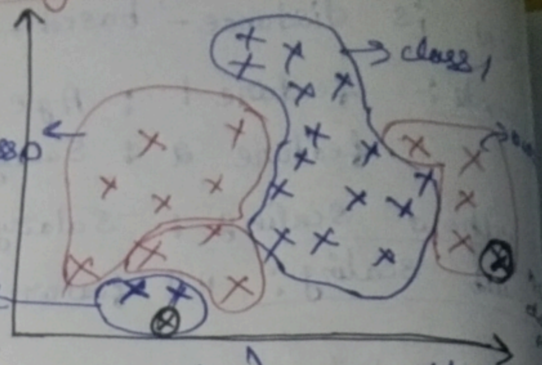
use mlxtend to plot surfaces

- A decision surface is the boundary that separates different class regions learned by a classifier

Overfitting and Underfitting in KNN :

cgpa	iq	placed
⋮	⋮	⋮

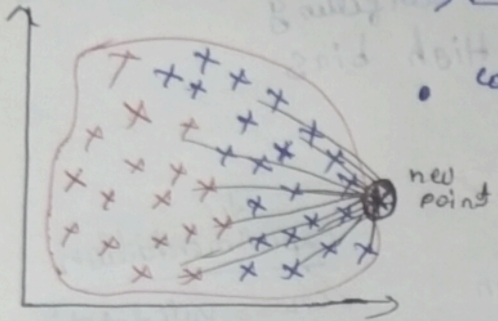
↕ 200 datapoints



i) $\boxed{\text{when } k = 1}$ $\boxed{\text{high variance}}$

→ if new data point is nearest to the red region (even if it is a an outlier), then it belongs to class 0.

→ $\boxed{\text{when } k = \text{maximum}}$, $\boxed{k = 200}$



- consider blue points are more in n… blue > red
- majority count ⇒ $\boxed{\text{class blue}}$ ⇒ minority patterns lost
- is bia…
- many far-away points influence prediction
- local patterns are ignored. (minority class are ignored) → cannot capture data pattern

Limitations of KNN:

i) Slow Prediction Time:- for every new data point, kn…
- computes distance to all training points
- Sorts them ascending
- Complexity ⇒ $\boxed{O(N \times D)}$

→ N = Number of training samples

→ D = number of features

- Not suitable for large datasets or real-time system.

ii) High Memory Usage :-
- KNN Stores the entire training dataset.
- No model Compression.

→ memory Inefficient for big data

iii) High dimensional data :- $\boxed{f = 500}$ / curse of dimensionality

- curse of dimensionality ⟶ distance concept ⇒ not reliable

iv) Large datasets :- n=5 lack , f=100 (5,00,000, 100)

- Knn is a lazy learning technique.
- training during → nothing

v) Sensitive to feature Scaling :-

- Distance based algorithm
- Features with large scale dominate

→ without scaling ⟹ wrong neighbors

→ Requires mandatory scaling

Ex :-

- age (0 - 100)
- Salary (0 - 10,00,000)

vi) Sensitive to noise and outliers :-

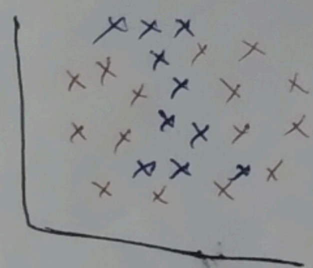
→ Nearby noisy points influence prediction

→ Especially when k is small.

→ A single noisy point can change classification.

vi) Poor Performance on imbalanced dataset :-

- majority class dominates voting , Minority class gets ignored . Bias predictions



→ Yes ⇒ 91%

→ No ⇒ 9%.

vii) No learning or Interpretability :-

- No parameters (weights, coefficients)
- No clear feature importance
- Hard to Explain predictions

viii) Affected by Irrelevant features :-

ix) No robust to missing values:

x) difficult to choos optimal k

xi)